REPORT

Data Laverager

ETL Project
Power BI Power Query
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INSTRUCTIONS

This project focuses on performing ETL (Extract, Transform, Load) operations using Power BI Power Query. The objective is to collect data from multiple sources, clean and transform it, and integrate it into a structured format suitable for analysis.

The project demonstrates how raw data can be converted into meaningful insights through proper data processing techniques. It includes handling multiple files, fixing data inconsistencies, and automating the workflow.

DATA SOURCES USED

Web Data (GDP Table)

- Extracted from an HTML source and used for additional data processing practice

Employee Dataset

- Contains employee details such as Employee_ID, Name, Department, Age , Date and Region

Sales Data (January, February, March, April Excel files)

- Contains transaction-level data including Revenue, Cost, and Order Date

TRANSFORMATIONS APPLIED

Data Cleaning

- Removed null values and blank rows to ensure data consistency
- Removed duplicate records to avoid incorrect analysis
- Fixed incorrect data types (Text to Number, Date formats)

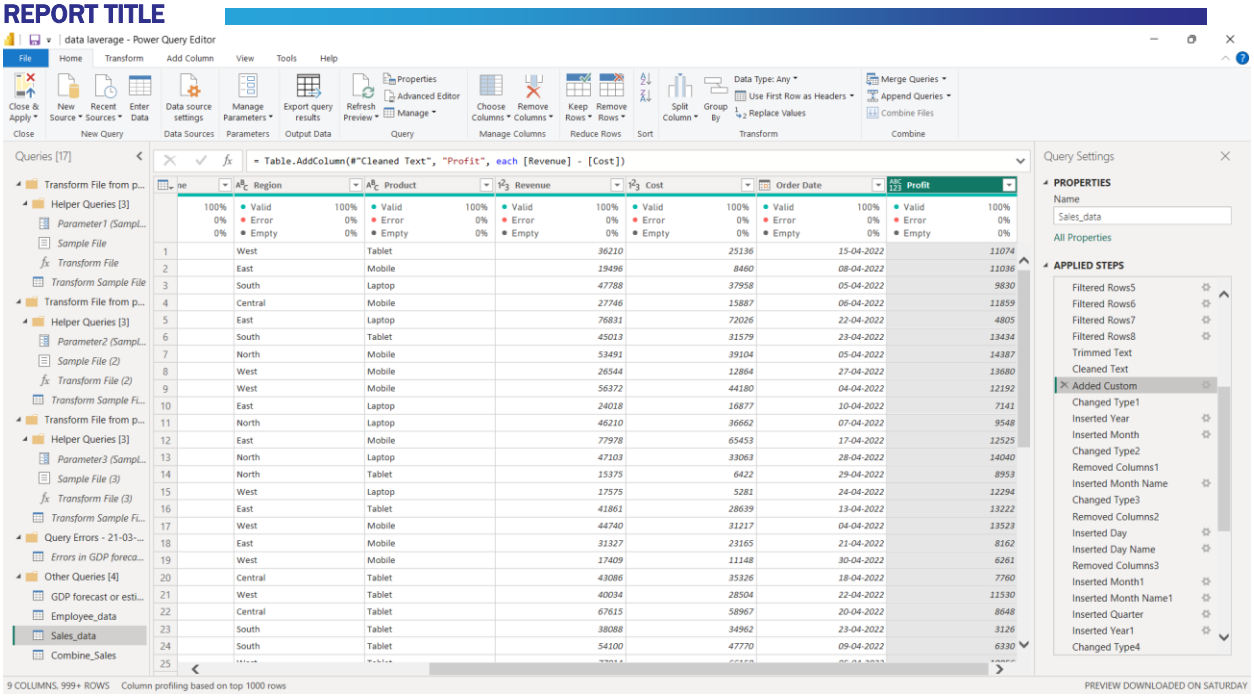
Text Transformation

- Used TRIM function to remove extra spaces.
- Used CLEAN function to remove hidden characters.
- Standardized text using proper case formatting

Numeric Transformation

- Created a Profit column using the formula:
  Profit = Revenue – Cost.

- Fixed numeric format issues in GDP data by:
  - Removing commas
  - Converting text into numeric values.



You Can See Process There

* Data Cleaning
* Text Transformation
* Numeric Transformation

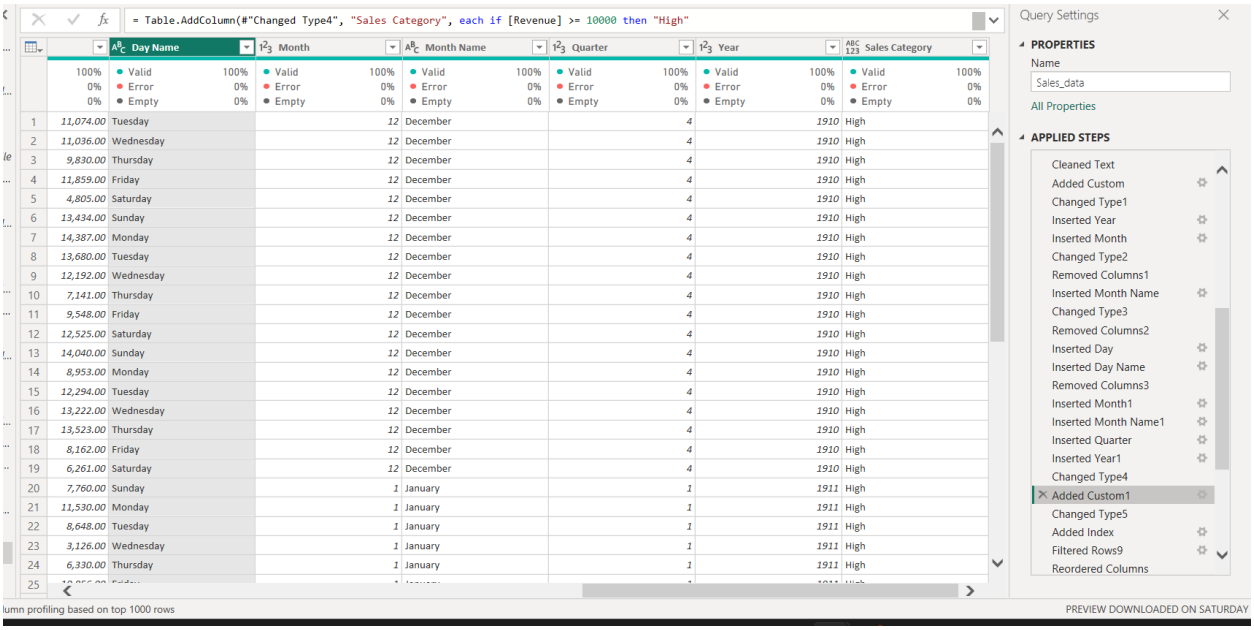
Conditional Column

• Created a Sales Category column based on Revenue:

- High (Revenue ≥ 10000)
- Medium (Revenue between 5000 and 9999)
- Low (Revenue < 5000)

Date Transformation

- Extracted Year, Month, and Quarter from Order Date.
- Enabled time-based analysis for better insights



	Day Name	Month	Month Name	Quarter	Year	Sales Category
	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty
1	11,074.00 Tuesday	12	December	4	1910	High
2	11,036.00 Wednesday	12	December	4	1910	High
3	9,830.00 Thursday	12	December	4	1910	High
4	11,859.00 Friday	12	December	4	1910	High
5	4,805.00 Saturday	12	December	4	1910	High
6	13,434.00 Sunday	12	December	4	1910	High
7	14,387.00 Monday	12	December	4	1910	High
8	13,680.00 Tuesday	12	December	4	1910	High
9	12,192.00 Wednesday	12	December	4	1910	High
10	7,141.00 Thursday	12	December	4	1910	High
11	9,548.00 Friday	12	December	4	1910	High
12	12,525.00 Saturday	12	December	4	1910	High
13	14,040.00 Sunday	12	December	4	1910	High
14	8,953.00 Monday	12	December	4	1910	High
15	12,294.00 Tuesday	12	December	4	1910	High
16	13,222.00 Wednesday	12	December	4	1910	High
17	13,523.00 Thursday	12	December	4	1910	High
18	8,162.00 Friday	12	December	4	1910	High
19	6,261.00 Saturday	12	December	4	1910	High
20	7,760.00 Sunday	1	January	1	1911	High
21	11,530.00 Monday	1	January	1	1911	High
22	8,648.00 Tuesday	1	January	1	1911	High
23	3,126.00 Wednesday	1	January	1	1911	High
24	6,330.00 Thursday	1	January	1	1911	High

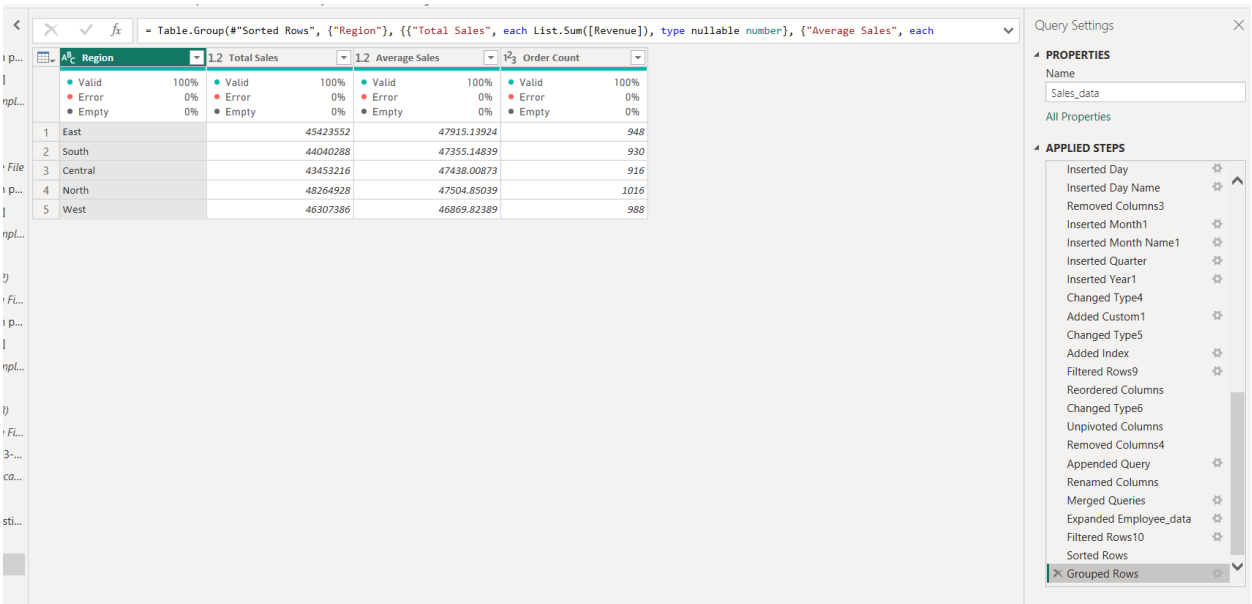
Data Integration

- Used Folder connector to append multiple monthly sales files.
- Merged Sales Data with Employee Data using EmployeeID.
- Ensured all records were retained using Left Join

Aggregation (Group By)

- Grouped data by Region to calculate:

- Total Sales (Sum of Revenue)
- Average Sales
- Number of Transactions



Data Profiling

- Used Column Quality, Column Distribution, and Column Profile tools.
- Identified errors, null values, and data inconsistencies.
- Ensured final dataset has 100% valid values (See In Previous Picture)

Automation

- Implemented folder-based data ingestion.
- Added a new file (April data) and used Refresh.
- Verified that data updates automatically without manual effort

CHALLENGES FACED

During the project, several challenges were encountered:

• Data type errors in GDP data due to different number formats
• Inconsistent text formatting (extra spaces and mixed cases)
• Managing and combining multiple Excel files efficiently
• Handling errors during transformation steps

SOLUTIONS

The following solutions were applied to overcome challenges

• Used Replace Values and Change Type with Locale for numeric issues
• Applied TRIM and CLEAN functions for text standardization
• Used Folder connector for automatic file combination
• Carefully debugged errors using data profiling tools

CONCLUSION

This project successfully demonstrates the implementation of an end-to-end ETL pipeline using Power BI Power Query.

The data was extracted from multiple sources, cleaned, transformed, and integrated into a structured format. The final dataset is accurate, consistent, and ready for analysis.

The project also highlights the importance of data quality, automation, and efficient data processing in real-world scenarios.

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Thank You